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Novel method for parameter estimation of Newton's rings based on CFRFT and ER-WCA

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ABSTRACT

Newton's rings is one of the important optical signals, it can be used to measure the curvature radius of optical lens. Owing to the photoelectronic digitized realization and newly established mathematical model of Newton's rings, the measuring of curvature radius can be changed to parameter estimation of Newton's rings. Therefore, some modern signal processing tools have been successfully applied to parameter estimation of Newton's rings, such as Fourier transform (FT), chirp-Fourier transform (CFT) and fractional Fourier transform (FRFT). Nowadays, it has been one of the latest developments in optical signal processing field. In this paper, a novel method for parameter estimation of Newton's rings is proposed based on concise fractional Fourier transform (CFRFT) and evaporation rate based water cycle algorithm (ER-WCA). A pretreatment technology is further proposed to solve uneven illumination of actual Newton's rings. The experimental results show that the efficiency of proposed method is much higher than that of FRFT-based method. It can also be seen from the experimental results that the proposed pretreatment technology can increase the accuracy.

1. Introduction

Interference fringe is one of important optical signals, it has been applied to many measurement fields like engineering surveying, spectrum analysis and radar [1]. The latest building measuring system based on computer and charge coupled device (CCD) realizes the measurement automation and the instrumental intelligence [2–6]. Various interference fringes have been designed for different purposes. Newton's rings is one of the important interference fringe. It is widely used in the optical metrology, such as testing spherical and flatting optical surface, determination of curvature radius, measurement of length and displacement [7]. The automatic measuring system of curvature radius of optical lens is shown in Fig. 1 (a). An actual digital Newton's rings is shown in Fig. 1 (b).

Since Newton's rings can be viewed as a 2-D multi-component chirp signal [8], the issue of measuring curvature radius can thus be changed to parameter estimation of 2-D multi-component chirp signal. The detection and parameter estimation of chirp signal have become one of the hottest topics in signal processing field [9–15]. Owing to the digitization and mathematical model of Newton's rings, some modern signal processing tools have been successfully applied to parameter estimation of Newton's rings, such as Fourier transform (FT) [16], chirp-Fourier transform (CFT) [17] and fractional Fourier transform (FRFT) [18,19]. These methods have advantages of robustness to noise and obstacle, but they still have some drawbacks like initial value dependence and long time consuming. For example, the FT-based method needs three iterations to isolate of one side band during fine determination of the fringe parameters, where iteration is time consuming; the FRFT-based method uses equal step length searching to obtain matched angle, speed and accuracy can not be guaranteed simultaneously; Nelder-Mead simplex search optimization using in CFT-based method depends on initial value and converges slowly.

Owing to above facts, a novel method for parameter estimation of Newton's rings is thus proposed based on concise fractional Fourier transform (CFRFT) and evaporation rate based water cycle algorithm (ER-WCA). It has the characteristics of less time consuming and independence on initial value owing to the following facts: (1) The CFRFT is a newly proposed signal processing tool [20]. It has the advantages of simple implementation and less computation compared with FRFT. (2) ER-WCA is a novel metaheuristic optimization algorithm [21]. It has been shown that in most cases the ER-WCA converges to the global solution faster and offers more accurate results than WCA and other considered optimizers, such as genetic algorithm (GA), particle swarm optimization (PSO), differ-

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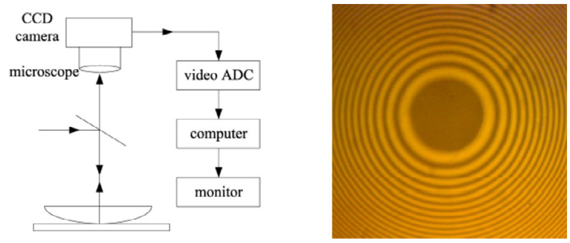


(a) (b)

Fig. 1. (a) Newton's rings experiment device, (b) Newton's rings image.

Table 1
Results obtained by using FRFT and proposed method for Newton's rings polluted by white Gaussian noise.

	FRFT		Proposed method		
SNR(db)	Error(%)	Time(s)	Radius(m)	Error(%)	Time(s)
10	0.22	981.7	0.8597	0.0349	24.1
5	0.22	990.9	0.8598	0.0233	23.9
0	0.22	983.2	0.8598	0.0233	24.1
−5	0.41	983.1	0.8594	0.0698	23.9
−10	1.3	999.0	0.8595	0.0581	24.0
−15	0.15	990.2	0.8590	0.1163	23.9
−20	34	992.6	0.8580	0.2326	24.2

ential evolution (DE) and artificial bee colony (ABC) [22]. In particularly, a new pretreatment technology is further proposed to solve the uneven illumination of actual Newton's rings, and thus effectively improve the measuring accuracy.

The main contributions of this paper are concluded as follows:

- We derive theoretically the frequency spectrum characteristic of Newton's rings in the CFRFT domain, and further propose a new optimization model to measure the curvature radius of optical lens based on this characteristic.
- ER-WCA is applied to solving this optimization model, and thus effectively improve the speed and accuracy.
- For practical, we propose two crucial technologies to reduce the impact of mean intensity and extend the dimensional normalization from 1-D to 2-D case.
- We propose a new pretreatment technology to solve the uneven illumination of actual Newton's rings.

The remaining sections of this paper are organized as follows: we review some basic concepts about CFRFT and mathematics model of Newton's rings in Section 2. In Section 3, a novel method of measuring curvature radius with Newton's rings is proposed using CFRFT and ER-WCA. The performance analysis of this method is shown in Section 4. Finally, conclusions are stated in Section 5.

2. Preliminaries

2.1. The concise fractional Fourier transform

The concise fractional Fourier transform (CFRFT) is a generation of the FT, its definition is shown as follows [20]:

$$F_\alpha(u) = F^\alpha[f(x)](u) = \int_R f(x)e^{i\pi(\cot\alpha x^2 - 2ux)}dx \tag{1}$$

where $\alpha \in [0, \pi]$, F^α and $F_\alpha(u)$ denote the CFRFT operator and CFRFT of $f(x)$ respectively.

The CFRFT of $f(x)$ can be calculated by chirp production and FT, i.e.,

$$F^\alpha[f(x)](u) = \int_R [f(x)e^{i\pi\cot\alpha x^2}]e^{-2i\pi ux}dx = F[f(x)e^{i\pi\cot\alpha x^2}] \tag{2}$$

where F denotes Fourier transform operator. For a N-point sequence, the computation of CFRFT is $N + N/2\log_2 N$. The computation of FRFT proposed in [23] is $8N + 6N\log_2(2N)$ [20]. Based on above analysis, CFRFT is easier to implement and has less computation than FRFT.

The 2-D CFRFT is defined as

$$\begin{aligned} F_\alpha(u, v) &= F^\alpha[f(x, y)](u, v) \\ &= \int_{R^2} f(x, y)e^{i\pi(\cot\alpha x^2 - 2ux)}e^{i\pi(\cot\alpha y^2 - 2vy)}dxdy \end{aligned} \tag{3}$$

where $\alpha \in [0, \pi]$, $F_\alpha(u, v)$ denotes the 2-D CFRFT of $f(x, y)$.

The Parseval principle associated with 2-D CFRFT can be obtained using the Parseval principle in Fourier domain, i.e., $\int_{R^2}|F_\alpha(u, v)|^2dudv = \int_{R^2}|f(x, y)|^2dxdy$. Informally, the identity asserts that the energy of CFRFT coefficients of signal is equal to the energy of signal, it can be proved briefly as following:

$$\begin{aligned} \int_{R^2}|F_\alpha(u, v)|^2dudv &= \int_{R^2}|F[f(x, y)e^{i\pi(\cot\alpha x^2 - 2ux)}e^{i\pi(\cot\alpha y^2 - 2vy)}]|^2dxdy \\ &= \int_{R^2}|f(x, y)e^{i\pi(\cot\alpha x^2 - 2ux)}e^{i\pi(\cot\alpha y^2 - 2vy)}|^2dxdy \\ &= \int_{R^2}|f(x, y)|^2dxdy \end{aligned} \tag{4}$$

2.2. Mathematical model of Newton's rings

Firstly, the chirp signal is briefly introduced. The chirp signal is a typical non-stationary signal, which is widely used in information systems such as radar, sonar, and communications. The definition of chirp signal is shown as follows:

$$g(x) = A \cdot rect(x/T)e^{i(\pi kx^2 + 2\pi f_0 x + \phi)} \tag{5}$$

where A is amplitude, T is time width, k is chirp rate, f_0 is center frequency and ϕ is phase. $rect(\cdot)$ is rectangular function.

Without the loss of generality, the mathematical model of Newton's rings is demonstrated as a 2-D multi-component chirp signal [8], i.e.,

$$\begin{aligned} f(x, y) &= I_0 - I_1 \cdot rect(x/T)rect(y/T)\cos\left[\pi\frac{2}{\lambda_0 R}(x - x_0)^2 \right. \\ &\quad \left. + \pi\frac{2}{\lambda_0 R}(y - y_0)^2\right] \\ &= I_0 - \frac{I_1}{2}\cdot rect(x/T)rect(y/T)\left(e^{i\pi\frac{2}{\lambda_0 R}[(x-x_0)^2 + (y-y_0)^2]}\right. \\ &\quad \left. + e^{-i\pi\frac{2}{\lambda_0 R}[(x-x_0)^2 + (y-y_0)^2]}\right) \end{aligned} \tag{6}$$

where I_0 represents the mean intensity, I_1 represents the amplitude of the fringes. T is the physical dimension of Newton's rings, R is the curvature radius of optical lens, λ_0 is the wavelength of the incident light and (x_0, y_0) denotes the coordinate of Newton's rings' center.

As shown above, Newton's rings can be viewed as the sum of three chirp signals, namely $f(x, y) = f_1(x, y) + f_2(x, y) + f_3(x, y)$, where $f_1(x, y) = I_0$, $f_2(x, y)$ and $f_3(x, y)$ are given as follows:

$$f_2(x, y) = -\frac{I_1}{2}\cdot rect(x/T)rect(y/T)e^{i\pi\frac{2}{\lambda_0 R}[(x-x_0)^2 + (y-y_0)^2]}$$

$$f_3(x, y) = -\frac{I_1}{2}\cdot rect(x/T)rect(y/T)e^{-i\pi\frac{2}{\lambda_0 R}[(x-x_0)^2 + (y-y_0)^2]}$$

The chirp rates of each component are $k_1 = 0$, $k_2 = \frac{2}{\lambda_0 R}$, $k_3 = -\frac{2}{\lambda_0 R}$ respectively.

3. Proposed method and its algorithm

In this section, the proposed method and its detailed algorithm are presented respectively in Section 3.1 and Section 3.2.

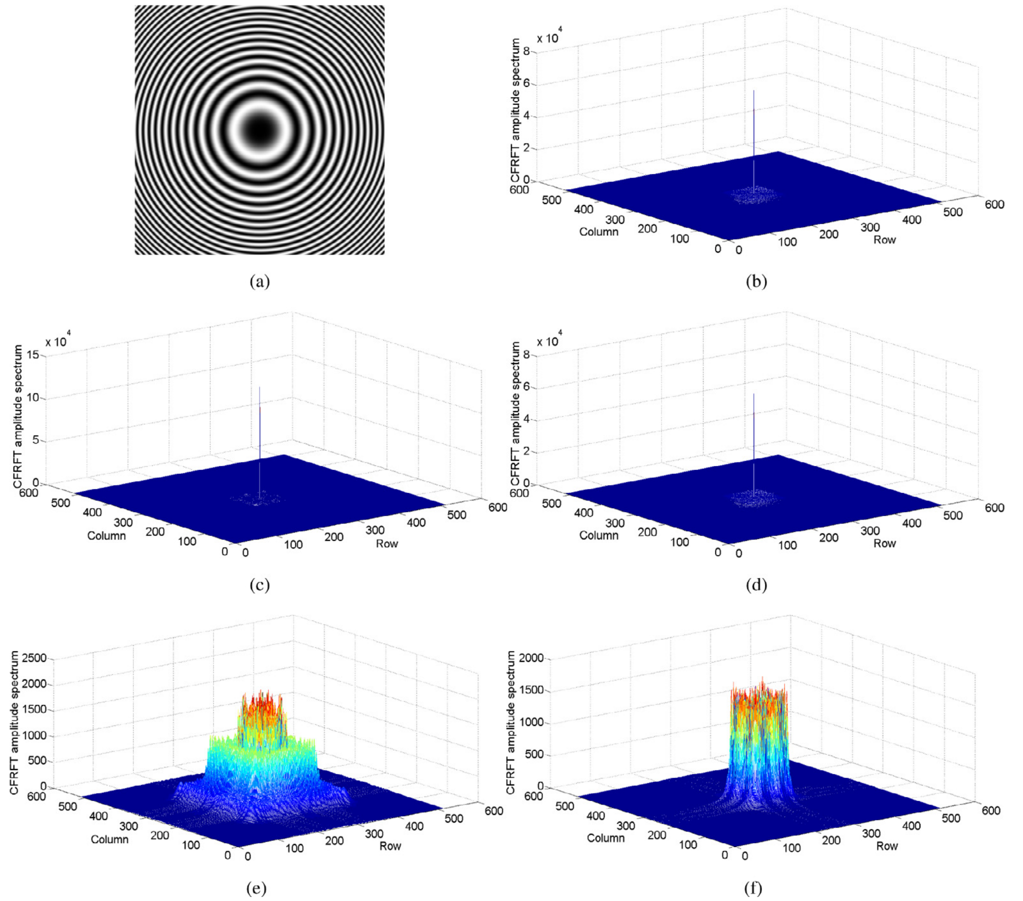


Fig. 2. The Newton's rings and CFRFT frequency spectrum with different angles (a) Newton's rings, (b) $\alpha = 0.8881\pi/2$, (c) $\alpha = \pi/2$, (d) $\alpha = 1.1119\pi/2$, (e) $\alpha = 1.2\pi/2$, (f) $\alpha = \pi/2$.

3.1. Proposed method

Based on the Eqs. (3) and (6), the frequency spectrum characteristic of Newton's rings in CFRFT domain is deduced as follows:

$$
\begin{aligned}
F_\alpha(u,v) &= F^\alpha[rect(x/T)rect(y/T)e^{i\pi[\frac{2}{\lambda_0 R}(x-x_0)^2+\frac{2}{\lambda_0 R}(y-y_0)^2]}](u,v)\\
&= \int_{R^2} rect(x/T)rect(y/T)e^{i\pi[\frac{2}{\lambda_0 R}(x-x_0)^2+\frac{2}{\lambda_0 R}(y-y_0)^2]}\\
&\quad e^{i\pi[\cot\alpha(x^2+y^2)-2(xu+vy)]}dxdy\\
&= \int_{-\frac{T}{2}}^{\frac{T}{2}}\int_{-\frac{T}{2}}^{\frac{T}{2}} e^{i\pi[(\frac{2}{\lambda_0 R}+\cot\alpha)x^2-2(u+\frac{2}{\lambda_0 R}x_0)x+\frac{2x_0{}^2}{\lambda_0 R}]}\\
&\quad e^{i\pi[(\frac{2}{\lambda_0 R}+\cot\alpha)y^2-2(v+\frac{2}{\lambda_0 R}y_0)y+\frac{2y_0{}^2}{\lambda_0 R}]}dxdy\\
&= e^{i\pi(\frac{2x_0{}^2}{\lambda_0 R}+\frac{2y_0{}^2}{\lambda_0 R})}\int_{-\frac{T}{2}}^{\frac{T}{2}}\int_{-\frac{T}{2}}^{\frac{T}{2}} e^{i\pi[(\frac{2}{\lambda_0 R}+\cot\alpha)x^2-2(u+\frac{2}{\lambda_0 R}x_0)x]}\\
&\quad e^{i\pi[(\frac{2}{\lambda_0 R}+\cot\alpha)y^2-2(v+\frac{2}{\lambda_0 R}y_0)y]}dxdy
\end{aligned}
$$

According to the characteristic of chirp signal, most energy of $F_\alpha(u,\ v)$ is concentrated at $[-\frac{2}{\lambda_0 R}x_0-(\frac{2}{\lambda_0 R}+\cot\alpha)T/2, -\frac{2}{\lambda_0 R}x_0+(\frac{2}{\lambda_0 R}+\cot\alpha)T/2]\times[-\frac{2}{\lambda_0 R}y_0-(\frac{2}{\lambda_0 R}+\cot\alpha)T/2, -\frac{2}{\lambda_0 R}y_0+(\frac{2}{\lambda_0 R}+\cot\alpha)T/2]$. Therefore, when $\cot\alpha = -\frac{2}{\lambda_0 R}$, then

$$
\begin{aligned}
F_\alpha(u,v) &= F^\alpha[e^{i\pi[\frac{2}{\lambda_0 R}(x-x_0)^2+\frac{2}{\lambda_0 R}(y-y_0)^2]}](u,v)\\
&= e^{i\pi(\frac{2x_0{}^2}{\lambda_0 R}+\frac{2y_0{}^2}{\lambda_0 R})}\int_{-\frac{T}{2}}^{\frac{T}{2}}\int_{-\frac{T}{2}}^{\frac{T}{2}} e^{-i\pi[2(u+\frac{2}{\lambda_0 R}x_0)x]}e^{-i\pi[2(v+\frac{2}{\lambda_0 R}y_0)y]}dxdy\\
&= e^{i\pi(\frac{2x_0{}^2}{\lambda_0 R}+\frac{2y_0{}^2}{\lambda_0 R})}\frac{\sin[\pi T(u+\frac{2}{\lambda_0 R}x_0)]}{2\pi(u+\frac{2}{\lambda_0 R}x_0)}\cdot\frac{\sin[2\pi T(v+\frac{2}{\lambda_0 R}y_0)]}{2\pi(v+\frac{2}{\lambda_0 R}y_0)}\\
&= \frac{T^2}{4}e^{i\pi\left(\frac{2x_0{}^2}{\lambda_0 R}+\frac{2y_0{}^2}{\lambda_0 R}\right)}sinc\Big[\pi T(u+\frac{2}{\lambda_0 R}x_0)\Big]\\
&\quad\cdot sinc\Big[\pi T(v+\frac{2}{\lambda_0 R}y_0)\Big] \qquad (7)
\end{aligned}
$$

where $sinc(\cdot)$ is sinc function.

Based on above theoretical derivation, the energy of $F_\alpha(u,\ v)$ concentrated at $(-\frac{2}{\lambda_0 R}x_0, -\frac{2}{\lambda_0 R}y_0)$ when $\cot\alpha = -\frac{2}{\lambda_0 R}$. According to the Parseval principle in CFRFT domain, the CFRFT frequency spectrum of $f(x,\ y)$ reach global maximum at $(-\frac{2}{\lambda_0 R}x_0, -\frac{2}{\lambda_0 R}y_0)$ when $\cot\alpha = -\frac{2}{\lambda_0 R}$. Similar to above deduction, we can conclude that the CFRFT frequency spectrum of Newton's rings can be seen obvious impulse with three matched angles, i.e., $\alpha_1 = \pi - \text{arccot}\frac{2}{\lambda_0 R}, \alpha_2 = \pi/2, \alpha_3 = \text{arccot}\frac{2}{\lambda_0 R}$.

Then, the above results are verified by Matlab. The Newton's rings is shown in Fig. 2 (a), its CFRFT frequency spectrum with different angles are shown in Figs. 2 (b–e), where (b–d) are obtained by using matched angles $\alpha_1 = 0.8881\pi/2, \alpha_2 = \pi/2, \alpha_3 = 1.1119\pi/2$ respectively, (e) is obtained by using unmatched angle $\alpha_4 = 1.2\pi/2$. From these figures, it can be seen that the CFRFT frequency spectrum of Fig. 2 (a) can be observed impulse with the matched angles, no impulse can be observed in Fig. 2 (e) with unmatched angle. This further verifies the correctness of Eq. (7) obtained by theoretical derivation. The Fig. 2 (f) is the CFRFT frequency of Newton's rings (see Fig. 2 (a)) subtract the mean intensity with matched angle $\alpha_2 = \pi/2$. Compared with Fig. 2 (c), no

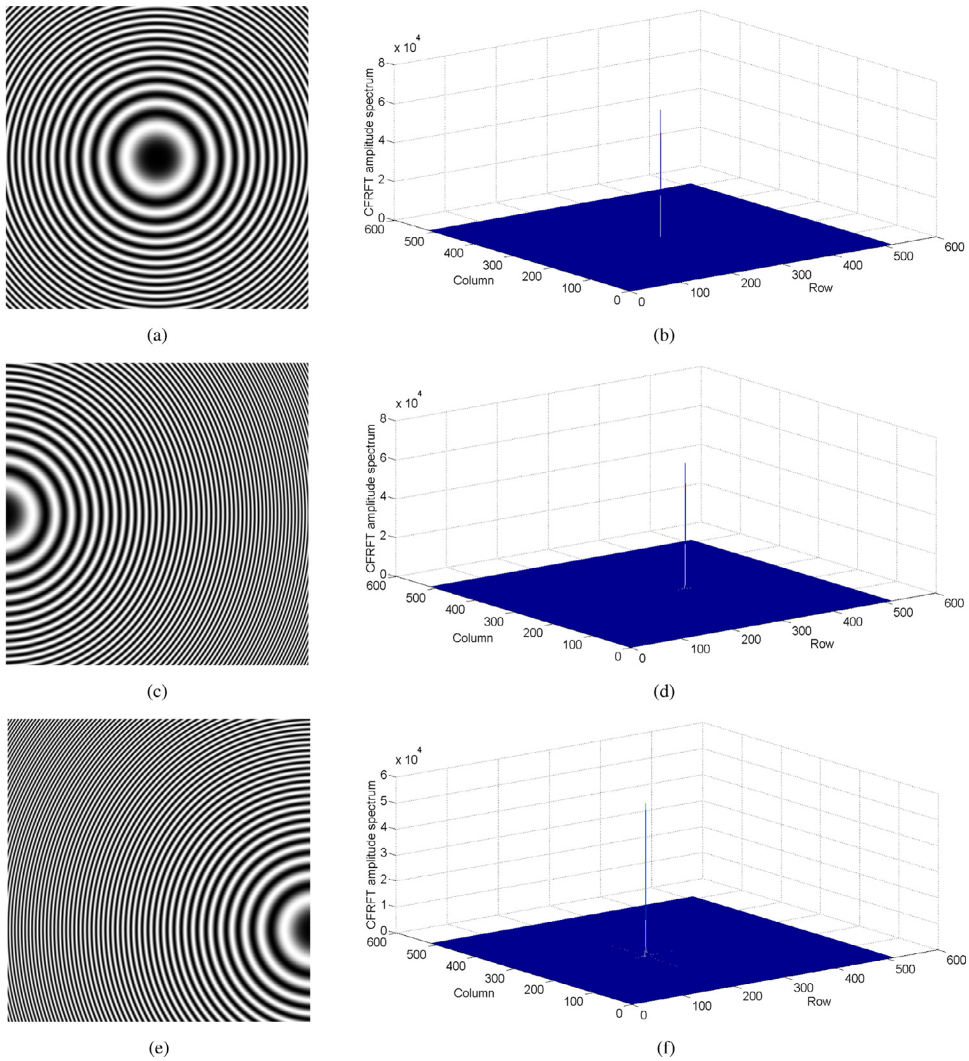


Fig. 3. The Newton's rings with different center and corresponding CFRFT frequency spectrum (a) (0,0), (b) CFRFT frequency spectrum of (a), (c) (−265,0), (d) CFRFT frequency spectrum of (c), (e) (265,100), (f) CFRFT frequency spectrum of (e).

impulse can be observed in Fig. 2 (f). Thus, it can be used to reduce the impact of mean intensity, which is useless in measuring curvature radius. This is one of the techniques in designing the algorithm of proposed method. In following of this paper, the CFRFT frequency spectrum of $f(x, y)$ means the CFRFT frequency spectrum of $f(x, y)$ subtract the mean intensity.

In view of this impulse characteristic with matched angle, the issue of measuring curvature radius with Newton's rings can be modeled as a optimization problem. Therefore, a new model is proposed to measure curvature radius with Newton's rings, i.e.,

$$\begin{cases} \alpha_0 = \arg \min\limits_{\pi/2 \le \alpha \le \pi} \{-\max |F^{\alpha}[f(x,y) - a](u,v)|\} \\ R_0 = -\frac{2}{\lambda_0 \cot \alpha_0} \end{cases} \tag{8}$$

where a is the derived value of mean intensity, α_0 is the obtained matched angle, λ_0 is the wavelength of the incident light and R_0 is the estimated curvature radius of optical lens.

In this paper, ER-WCA is used to solve this optimization model. The advantages of ER-WCA are that it converges to the global solution faster and offers more accurate results than WCA and other considered optimizers [21]. The water cycle algorithm is a novel metaheuristic optimization algorithm proposed by Eskandar et al. [22]. It is inspired by the observation of water cycle process and how rivers and streams flow to the sea. ER-WCA was modified using the new concept of evaporation rate for different rivers and streams. The detail algorithm can be found in [21].

3.2. Algorithm

For a Newton's rings, suppose its size is $N \times N$, physical dimension is Tmm $\times$ Tmm. The algorithms of proposed method are shown as following:

Step 1. Read the Newton's rings image into computer and transform its gray value to [0,1], denoted as $f(m, n)$, where m, n are positive integers and $1 \le m, n \le N$.

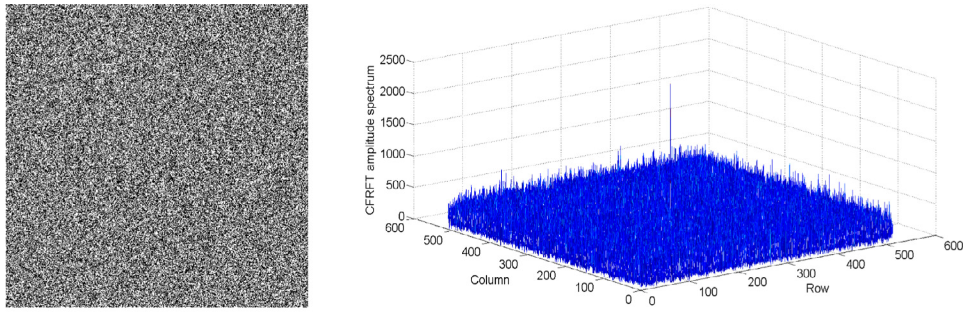
Step 2. $f(m, n)$ minus the derived value of mean intensity, then multiplied by a chirp signal as following:

$$g(m,n) = [f(m,n) - a] e^{i\pi \cot\alpha (m/f_s')^2} \cdot e^{i\pi \cot\alpha (n/f_s')^2} \tag{9}$$

where $f_s' = \sqrt{N}$, $a = 1/2, \alpha \in [\pi/2, \pi]$.

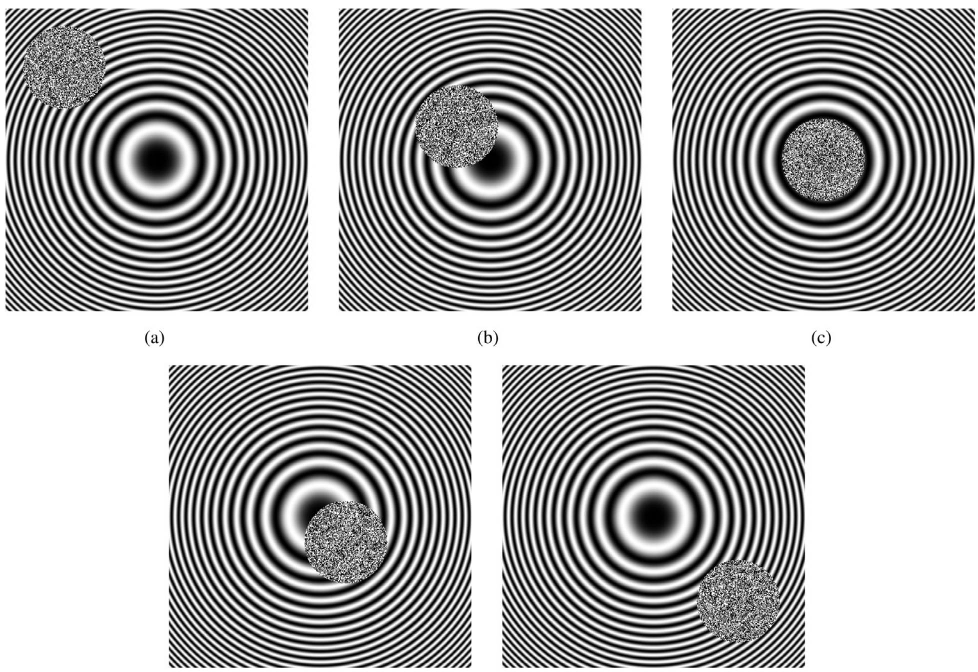
Two key technologies used in this step are described in *Remark 1* and *Remark 2*:

Remark 1. The dimensional normalization technology is frequently used in the preprocessing of discrete 1-D signal [23], we extend this technology to 2-D case. The factor with time dimensional is introduced as $S = \sqrt{T/f_s}$, $f_s = N/T$. A new coordinate system is defined as: $(x, y) \to (x/S, y/S)$, $(u, v) \to (u \cdot S, v \cdot S)$. Based on



(a) (b)

Fig. 4. (a) The polluted Newton's rings (SNR=−20 db), (b) CFRFT frequency spectrum of (a).



(a) (b) (c) (d) (e)

Fig. 5. The obscured Newton's rings (a) (100,100), (b) (200,200), (c) (256,256), (d) (300,300), (e) (400,400).

Table 2
Results obtained by using FRFT and proposed method for obscured Newton's rings.

	FRFT		Proposed method		
Obstacle(pixel)	Error(%)	Time(s)	Radius(m)	Error(%)	Time(s)
(100,100)	0.41	990.4	0.8597	0.0349	24.0
(200,200)	0.41	982.5	0.8598	0.0233	24.1
(256,256)	0.22	976.7	0.8601	0.0116	24.7
(300,300)	0.41	980.0	0.8599	0.0116	24.0
(400,400)	0.41	983.8	0.8597	0.0349	24.2

Table 3
Results obtained by using FRFT and proposed method for polluted Newton's rings.

	Description			Results		
Fig. 6	Obstacle(pixel)	$(f_{0,x}, f_{0,y})$	Noise(db)	Radius(m)	Error(%)	Time(s)
(a)	(256,256)	(0,0)	5	0.86	0	24.0
(c)	(256,256)	(0,0)	−5	0.8602	0.0233	24.2
(e)	None	(−265,0)	−5	0.8603	0.0309	24.0
(g)	None	(265,100)	−5	0.8594	0.0689	24.2

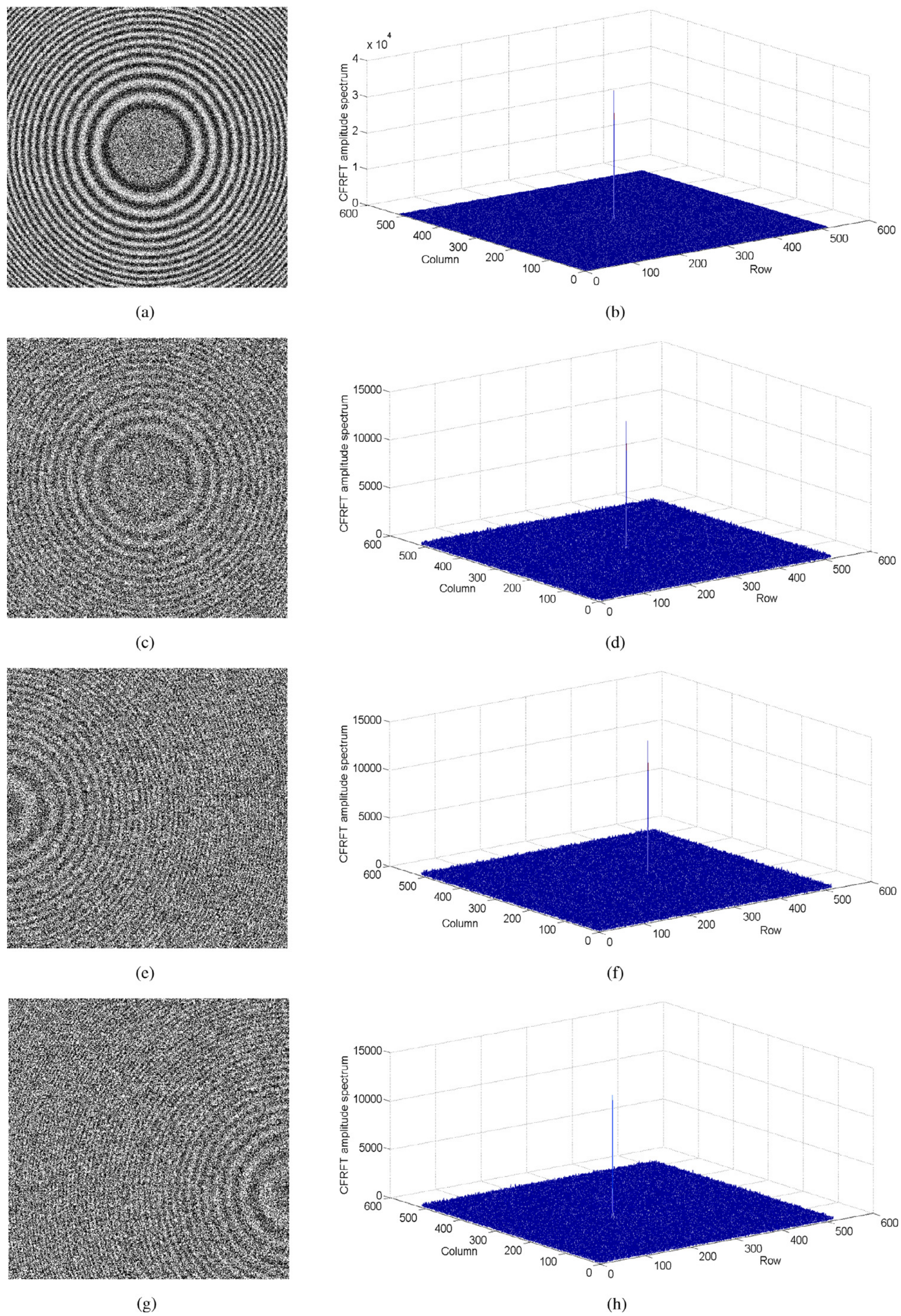


Fig. 6. The polluted Newton's rings and corresponding CFRFT frequency spectrum (a) noise(SNR=5 db)+obstacle, (b) CFRFT frequency spectrum of (a), (c) noise(SNR=−5 db)+obstacle, (d) CFRFT frequency spectrum of (c), (e) noise(SNR=−5 db), (f) CFRFT frequency spectrum of (e), (g) noise(SNR=−5 db), (h) CFRFT frequency spectrum of (g).

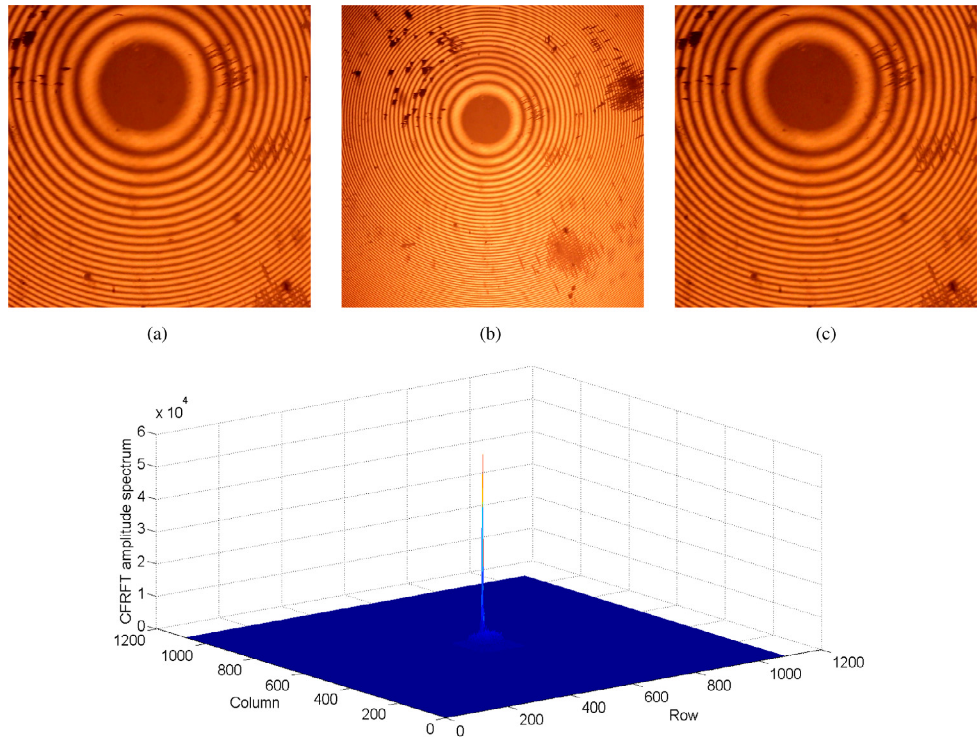
the new coordinate system, both time and frequency domain are $[-f'_s/2, f'_s/2] \times [-f'_s/2, f'_s/2]$, where $f'_s = \sqrt{T \cdot f_s} = \sqrt{N}$.

Remark 2. The selection of $a = 1/2$ is obtained by theoretical derivation. For pure Newton's rings (i.e. no noise, no obscure and uniform illumination), the range of $f(x, y)$ in Eq. (6) is $[I_0 - I_1, I_0 + I_1]$. And also because $mean(\cos x)$ is equal to zero in a cycle of 2π, so the mean value of this Newton's rings is about $I_0 = \frac{(I_0 - I_1) + (I_0 + I_1)}{2}$.

Step 3. fft2 is applied on $g(m, n)$ to obtain its amplitude spectrum, denoted as $G_\alpha(p, q) = |\mathrm{fft2}(g(m, n))|$, where p, q are positive integers and $1 \le p,\ q \le N$. fft2 denotes the 2-D fast Fourier transform.

Step 4. Compute the maximum value of $G_\alpha(p,\ q)$ and take $-\max(G_\alpha(p, q))$ as the fitness function of ER-WCA, optimized variable is α.

Step 5. Input initialization parameters of ER-WCA, including lower and upper bound of optimized variable LB and UB, number of design variables n_{vars}, population size N_{pop}, number of rivers + sea N_{sr}, evaporation condition constant d_{max} and maximum number of iterations N_{max}.



(d)

Fig. 7. The test actual Newton's rings (a) 720 × 720, (b) 1200 × 1200, (c) 1080 × 1080, (d) 3-D view of CFRFT frequency spectrum of (c).

Table 4
The estimated results for actual Newton's rings.

	Proposed method			Proposed method+pretreatment		
size	Radius(m)	Error(%)	Time(s)	Radius(m)	Error(%)	Time(s)
720 × 720	0.8559	0.4767	52.1	0.8576	0.2791	67.0
1080 × 1080	0.8414	2.1628	114.8	0.8524	0.8837	151.1
1200 × 1200	0.8614	0.1628	150.3	0.8611	0.1279	189.9

Step 6. Run the main program of ER-WCA to obtain the global optimal value α_0, i.e.,

$$\alpha_0 = \arg \min_{\pi/2 \le \alpha \le \pi} \{-|\max(G_\alpha(p,q))|\} \tag{10}$$

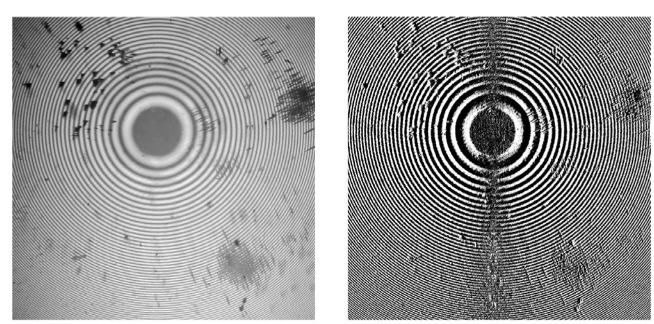
Step 7. Since the dimensional normalization using in step 2, the estimated curvature radius is finally obtained by the modified equation as follows:

$$R_0 = -\frac{2S^2}{\lambda_0 \cot \alpha_0} \tag{11}$$

where R_0 denotes the estimated curvature radius of optical lens, S is the factor with time dimensional and λ_0 is the wavelength of the incident light.

4. Experiment results and analysis

In order to verify the effectiveness of proposed method, the experiment and analysis are performed on the pure Newton's rings, polluted Newton's rings and actual Newton's rings. All experiments are conducted on a personal computer having Intel dual core 3.2 GHz CPU with 4.0 GB RAM, and using MATLAB version R2014a under the Windows 7 environment. The initialization parameters of ER-WCA are $(LB, UB, n_{vars}, N_{pop}, N_{sr}, d_{max}, N_{max}) = (\pi/2, \pi, 1, 50, 4, 10^{-26}, 20)$.



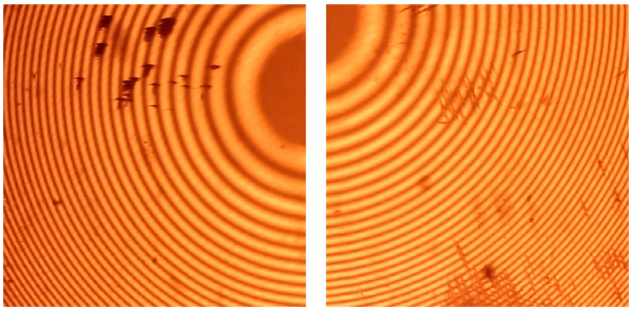
(a) (b)

Fig. 8. The gray scale of Newton's rings and pretreated Newton's rings (a) 1200 × 1200, (b) 1199 × 1199.

4.1. Analysis of pure Newton's rings

The discrete pure Newton's rings of size $N \times N$ is generated by using the following equation:

$$f(m,n) = I_0 - I_1 \cos[\pi \frac{2}{\lambda_0 R}(m/f_s - f_{0,x}/f_s)^2 + \pi \frac{2}{\lambda_0 R}(n/f_s - f_{0,y}/f_s)^2] \tag{12}$$



(a) (b)

Fig. 9. The actual Newton's rings whose center is out of sight (a) 512 × 512, (b) 512 × 512.

Table 5
Results for actual Newton's rings with some center errors.

Size	Center error(pixel)	Radius(m)	Error(%)	Time(s)
720 × 720	1	0.8576	0.2791	69.3
	3	0.8576	0.2791	70.7
	5	0.8576	0.2791	71.8
	7	0.8576	0.2791	70.2
1080 × 1080	1	0.8524	0.8837	153.1
	3	0.8524	0.8837	153.0
	5	0.8524	0.8837	163.5
	7	0.8524	0.8837	153.2
1200 × 1200	1	0.8611	0.1279	198.1
	3	0.8611	0.1279	197.2
	5	0.8611	0.1279	207.2
	7	0.8611	0.1279	192.4

where $m, n \in [-N/2, N/2-1]$, $f_s = N/T$ denotes the sampling frequency, T is the physical dimension of Newton's rings. $(f_{0,x}, f_{0,y})$ denotes the coordinate of Newton's rings' center.

In follows, the discrete 512 × 512 pure Newton's rings is generated by using $I_0 = I_1 = 2, N = 512$, $R = 0.86$ m, $\lambda_0 = 586.3$ nm, $T = 4.8$ mm, where λ_0 is the wavelength of sodium yellow lamp. When $f_{0,x} = f_{0,y} = 0$, the discrete pure Newton's rings is shown in Fig. 3 (a). When $f_{0,x} = -265, f_{0,y} = 0$, the center of this pure Newton's rings is out of sight (see Fig. 3 (c)). When $f_{0,x} = 265, f_{0,y} = 100$, the center of this pure Newton's rings is also out of sight(see Fig. 3 (e)).

For Figs. 3 (a), (c) and (e), the estimated curvature radii are 0.8597 m, 0.86 m and 0.8601 m, the processing time is about 24 s. In particularly, since traditional Newton's rings method is based on the physical characteristic of Newton's rings, i.e., the radius of each ring, it is not able to measure the curvature radius with Newton's rings whose center is out of sight (see Figs. 3 (c) and (e)). However, our method still has ability to measure the curvature radius. Based on the angle obtained by proposed method, the CFRFT frequency spectrum are shown respectively in Figs. 3 (b), (d) and (f). The validity of proposed method is also verified by the obvious impulse in Figs. 3 (b), (d) and (f).

4.2. Analysis of polluted Newton's rings

In Section 4.1, the ability of our method in measuring the curvature radius with pure Newton's rings is discussed. But pure Newton's rings is hard to obtained in reality, optical instruments and human operation inevitably cause image pollution. Therefore, the ability of our method in processing polluted Newton's rings is detected from three points. The specific are depicted as follows: (1) Newton's rings is polluted by white Gaussian noise; (2) Newton's rings is polluted by obstacle; (3) Newton's rings is polluted by both white Gaussian noise and obstacle.

The white Gaussian noise is a basic noise model used in information theory to mimic the effect of many random processes that occur in nature. Thus, for Newton's rings polluted by different white Gaussian noise, the ability of proposed method in measuring the curvature radius is detected firstly. The obtained results are listed in Table 1. Compared with the similar method based on FRFT, the proposed method is not only more accurate but also much faster. The great improvement of processing speed will be of great benefit to real-time measurement. In order to reflect the advantage of our method more intuitively, the Newton's rings polluted by white Gaussian noise with SNR=−20 db is shown in Fig. 4 (a). In this figure, the Newton's rings has been completely covered by white Gaussian noise. The proposed method can still obtain curvature radius with low error.

Then, for five Newton's rings (see Fig. 5) obscured by circular obstacle with same radius 100 pixels and different center positions, the obtained results are listed in Table 2. Compared with the results of pure Newton's rings, obstacle has very small impact on the estimated curvature radius. It is necessary to note that our method has great advantage than traditional Newton's ring methods in processing Newton's rings whose center are covered by obstacle (see Figs. 5 (c) and (d)). In comparison with the method based on FRFT, same conclusions can be derived like white Gaussian noise.

Since actual Newton's rings may includes not only one pollution source, the following discussions are divided into two parts: (1) the center of Newton's rings is visible, meanwhile it is polluted by Gaussian white noise and obstacle (see Figs. 6 (a) and (c)); (2) the center of Newton's rings is out of sight, meanwhile it is polluted by Gaussian white noise (see Figs. 6 (e) and (g)). All of above obtained results are listed in Table 3. Under these situations, the proposed method still has ability to estimate curvature radius with less error. Similar to pure Newton's rings, the obvious impulse can be seen from each CFRFT frequency spectrum (see Figs. 6 (b),(d),(f) and (h)).

4.3. Analysis of the actual Newton's rings

In this subsection, the actual Newton's rings are used to estimate curvature radius. As shown in Fig. 1 (a), the curvature radius of optical lens is 0.86 m and the wavelength of sodium lamp is 589.3 nm. The resolution of camera are 1600 × 1200 pixels (pixel size of 6.4 μm × 6.4 μm), 1920 × 1080 pixels (pixel size of 3.9 μm × 3.9 μm), and 1280 × 720 pixels (pixel size of 5.9 μm × 5.9 μm) respectively. For the convenience of experiment, these three figures are sheared to resolution of 720 × 720, 1200 × 1200 and 1080 × 1080 (see Figs. 7 (a–c). The center positions of three figures are (324, 192), (576, 452) and (487, 289) respectively.

For an actual Newton's rings whose center is visible, the uneven illumination may affect the experimental results. Therefore, a pretreatment technology is further applied on actual Newton's rings to solve the uneven illumination. Suppose $f(m, n)(1 \le m, n \le N)$ is the Newton's rings, then $f'(m, n)$ $(1 \le m, n \le N-1)$ is obtained by following equation:

$$f'(m,n) = \begin{cases} f(m,n) - f(m,n+1) & \text{if} \quad n \le x_0 \\ f(m,n+1) - f(m,n) & \text{if} \quad n > x_0 \end{cases} \tag{13}$$

where (x_0, y_0) is the center position of $f(m, n)(1 \le m, n \le N)$. Since each column or row of Newton's rings $f(m, n)(1 \le m, n \le N)$ is a 1-D chirp signal with same chirp rate, $f'(m, n)$ can also be viewed as a chirp signal, its chirp rate is identical with $f(m, n)$. The gray scale image of Fig. 7 (b) is shown in Fig. 8 (a). It can be seen that the brightness of those pixels closer to the center is brighter than those pixels in the four corners. However, the Fig. 8 (b) obtained by using Eq. (13) is brightness uniformity. It means that this technology can effectively solve the uneven illumination of actual Newton's rings.

Therefore, based on above elaborates, the results obtained by the proposed method and the proposed method combined with pretreatment technology are respectively listed in Table 4. By analyzing the data in this table, we can derive that the proposed method has good performance in estimating the curvature radius with actual Newton's rings. The curvature radius is obtained with an error of approximately 0.1628%, 2.1628% and 0.4767%. The errors obtained by using FRFT are about 0.34%, 0.9% and 1.27%, the errors of our method are less than the FRFT based method for 1200×1200 and 720×720. In further, the pretreatment technology can effectively decrease the errors in comparison with the method without using the pretreatment technology.

Based on above analysis, our proposed method combined with the pretreatment technology is fully superior FRFT-based method. Since the pretreatment technology is proposed by using the position of Newton's rings' center, it is necessary to verify that the results obtained by using the pretreatment technology is whether stable if the position of center has some pixels error. For Fig. 7 (a–c), suppose the position of center has 7 pixels error at most. The corresponding results obtained by using the propose method and the pretreatment technology are listed in the Table 5. By comparing these data with the data in Table 4, the estimated curvature radius are still stable when the position of center has 7 pixels error at most.

In order to test the ability of our method in processing the Newton's rings whose center is out of sight, two 512×512 images (see Fig. 9 (d and e)) are sheared from 1600×1200 image. The estimated curvature radius are respectively 0.8579 m and 0.8573 m, time consuming are 28.1 s and 28.0 s. This further illustrates the advantage of our method in processing Newton's rings whose center is out of sight.

5. Conclusions

In this paper, a novel method for parameter estimation of Newton's rings is proposed based CFRFT and ER-WCA, which can be further used to measure curvature radius of optical lens. This method combines the advantages of CFRFT and ER-WCA by modeling the issue of measuring curvature radius as a optimization problem. The efficiency of proposed method is much higher than FRFT-based method because CFRFT possesses the advantages of simple implementation and less computation, and ER-WCA has the advantages of fast convergence and high precision. In the design of algorithms, the two proposed technologies (see Remark 1 and Remark 2) reduce the impact of mean intensity and extend the dimensional normalization to 2-D case. The further proposed pretreatment technology can solve the uneven illumination of actual Newton's rings, and thus improve the accuracy.

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